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Summary of C Preprocessor

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C Preprocessor

Understanding the Preprocessor

In C programming, a preprocessor is utilized as a text-editing tool that runs on your source code prior to compilation occurring. Main tasks and primary functions encompass expanding macros, incorporating external files, and managing conditional compilation blocks. It processes and scans the source file for all lines beginning with a pound sign (#) for specific instructions for the preprocessor. Then after finding the instructions, it modifies the raw text based on those commands.

The C preprocessor (CPP) does not analyze syntax, understand data types, or process runtime logic.

Every preprocessor directive must start with a hash character (#), and unlike common practice C statements, do not end with a semicolon.

C Preprocessor : What it does

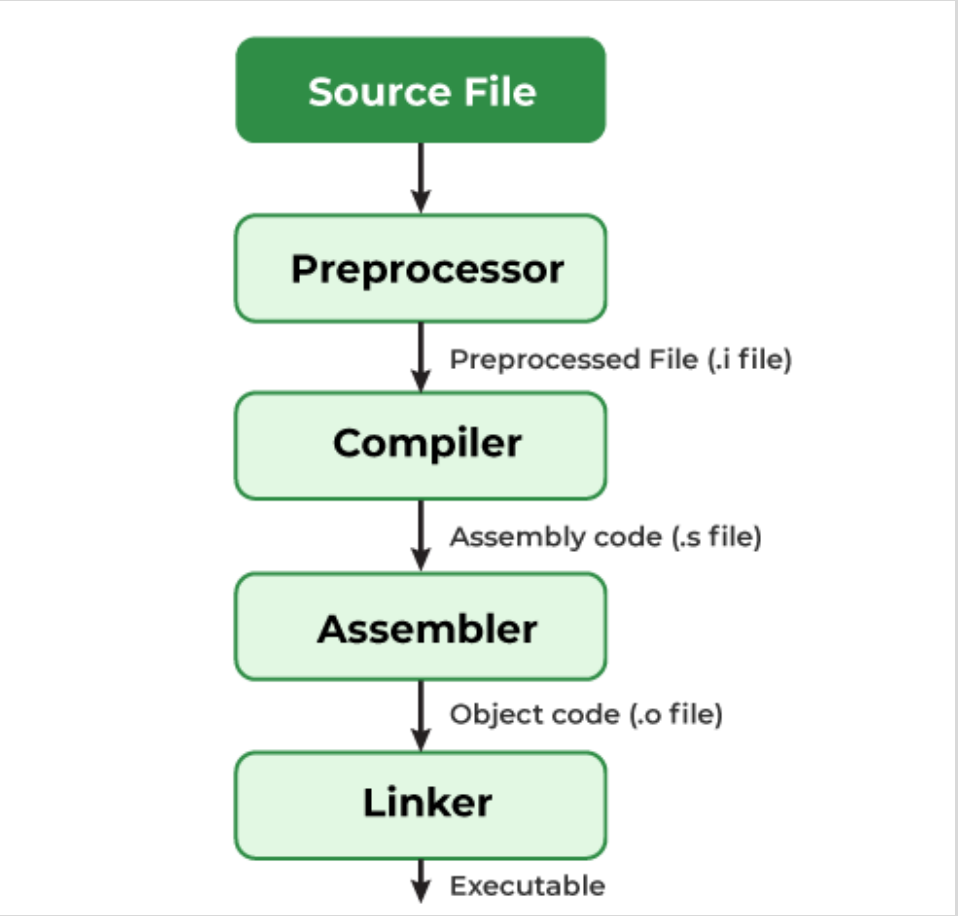
The C preprocessor (CPP) alters your source code before the compiler receives it. The preprocessor functions as the initial stage of the overall C compilation pipeline.

The preprocessor searches for specialized command lines commonly known as directives, which are designated by a leading hash character (#). Since the preprocessor focuses entirely on basic text manipulation without any knowledge of C syntax, data types, or structural grammar, it purely reorganizes and updates the raw source code. This output results in a fully expanded source file ready for compilation.

The C preprocessor that runs prior to actual compilation and operates as a distinct text-processing step. It is not part of the compiler itself although it is triggered automatically by the compiler to prep and set up the base of the code.

C Preprocessor : How it works

The diagram illustrates the chronological stages of the C compilation pipeline, tracking how a raw program transforms into a runnable application. The process begins with the initial source files, which passes directly into the preprocessor to yield an expanded preprocessed file. Next, the compiler translates this text into low-level architectural instructions, outputting an assembly code. The assembler then converts these instructions into machine-readable object code. Finally, the linker combines this object code with external libraries to build the final standalone executable.



C Preprocessors – GeeksforGeeks

C Preprocessor : Macros

Discuss macros definitions

Definition: A named code fragment managed by the preprocessor.

Purpose: Performed as a purely textual substitution before compilation begins.

When the preprocessor finds the macro name in your code and swaps it with the defined replacement text before handing the file to the compiler. When it comes to syntax rules, macro directives always begin with a hash symbol (#) and never end with a semicolon (;).

C Preprocessor : Conditional Compilation

Conditional compilation allows the C preprocessors to selectively shape the raw source text prior to compilation. It relies on macros existence and constant expressions to include or isolate specific blocks of code. Unlike runtime conditional logic, this evaluation happens strictly at compile time. Completely purging unused code from the executable binary to optimize size and efficiency.

C Preprocessor : Preprocessing directives

Preprocessing directives are commands that tell the C preprocessor how to handle source code before the compilation process begins. The #include directive allows header files to be added to a program, while #define creates macros and symbolic constants. Directives such as #ifdef, #ifndef, #if, #elif, #else, and #endif are used to control which sections of code are processed during compilation. The #undef directive removes a previously defined macro. These directives help programmers organize their code and control how their programs are prepared for compilation.

Sources

C Preprocessors – GeeksforGeeks

Preprocessor | Microsoft Learn

The C Preprocessor - The C Preprocessor

Preprocessors in C

The C Preprocessor's Arithmetic World - Developer Help

C Preprocessor and Macros

How a Preprocessor works in C? – GeeksforGeeks

C Preprocessors: Types & Workflow Guide

#30: Preprocesor and Macros | C Programming for Beginners

C/C++ Preprocessor Macros - DEV Community

Conditionals (The C Preprocessor)

#if #elif #else Conditional Compilation Directives | C Programming Tutorial

The C Preprocessor
describe macros related to C preprocessor in C programming – Search

https://learn.microsoft.com/en-us/cpp/preprocessor/preprocessor-directives?view=msvc-170

https://learn.microsoft.com/en-us/cpp/c-language/directives-to-the-preprocessor?view=msvc-170

https://learn.microsoft.com/en-us/cpp/preprocessor/hash-ifdef-and-hash-ifndef-directives-c-cpp?view=msvc-170

https://gcc.gnu.org/onlinedocs/cpp/

Conclusion

In conclusion, the C preprocessor is an important part of the C programming process because it prepares source code before it reaches the compiler. It performs tasks such as including header files, replacing macros, and controlling which sections of code are processed. Understanding preprocessing directives helps programmers better understand how their code is handled before compilation. Overall, the preprocessor plays an important role in organizing code and managing the compilation process.